STAT 340 Applied Methods in Statistics Course: Lecture 1 - Introduction

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STAT 340 - Introduction

Applied Methods in Statistics

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Research interests (what are yours?):

- Bayesian statistics and machine learning;
- Latent binary Bayesian neural networks;
- Bayesian generalized nonlinear models;
- Advanced mode jumping samplers and other approximation schemes for Bayesian inference.

Class Schedule

Lectures

- **When:** Tuesdays 14:15–16:00
- **Where:** Room T330
- **First Lecture:** Feb 3

Exercises (individual work)

- **When:** Thursdays 14:15–16:00
- **Where:** Room BT3A10
- **First Session:** Feb 5

Exercises (solutions presentation)

- **When:** Mondays 10:15–12:00
- **Where:** Room T132
- **First Session:** Feb 9

Practical Information

Mandatory Activity

- **Four compulsory quiz assignments**
- You can be released from a quiz by answering 10 questions correctly during lectures (no penalty for wrong answers).
- This will work even after failing a quiz.

Exam

- **Format:** Multiple choice (100%)
- **Grading:** A–E or Not Passed
- **Allowed:** All calculators and specified **written** aids
- **Date & Time: May 18, 09:00–12:00**
- **Important:** No date changes possible

Learning Activities (Weekly)

- **Lectures:** 2 hours
- **Individual Work:** 2 hours
- **Solutions:** up to 2 hours

Total: 6 hours per week

Assumed Background: STAT 100

Students should be familiar with:

- Basic rules of probability (we shall repeat that today)
- Common probability distributions (we shall repeat that today):
  - Binomial
  - Gaussian (Normal)
  - Student's t
  - Fisher's F
- Hypothesis testing:
  - One-sample t-tests
  - Two-sample t-tests
  - Paired sample t-tests
- One-way Analysis of Variance (ANOVA)
- Simple linear regression (one predictor variable)
- Contingency tables

Software & Data

Software

- **R Studio** for R programming language
- Practical analysis using various statistical methods
- Theoretical and practical interpretation of results
- (Other software may be used if methods are available)

Data

- Mainly R library: **BIAS.data** (GitHub)
  https://github.com/thoree/BIAS.data

STAT 340 Course Content

Main Topics

1. **Multiple Linear Regression**
2. **Multivariate Analysis**
   - Principal Component Analysis (PCA)
3. **Classification Methods**
4. **Clustering Methods**
5. **Generalized Linear Models (GLM)**
   - Binomial/Bernoulli (logistic) regression
   - Poisson regression
   - Mixed models (GLMM): Random effects, repeated measurements

Multiple Linear Regression

Principal Component Analysis

Classification Methods

Clustering Methods

Generalized Linear Models

Mixed Effect Models

Required Learning Resources

Primary Materials

- **Lecture presentations** (provided)
- **All exercises** (provided)

Recommended Textbooks (available online)

Main Reference: - **R Book:** Michael J Crawley, Wiley (2nd ed.)

- Chapters: 1–9 (as needed), 10–13, 17, 19, 25
- Freely available https://www.cs.upc.edu/~robert/teaching/estadistica/TheRBook.pdf

Supplementary (mathematical details): - **ISL:** James et al. "An Introduction to Statistical Learning' '

- Chapters: 1–4, 6.1, 7.1, 10
- Freely available https://www.statlearning.com

Response

Predictors	"Study of associations"	Continuous		Discrete		
		Normal	Non-normal	Counts	Bivariate	Multivariate
		"Linear"	(Transform?)	Poisson	Logistic	Multinomial
Fixed	Contin.	Linear Regression	GLM regression	GLM Regression	GLM regression or Discriminant analysis (LDA, QDA ++)	
Fixed	Discrete	ANOVA	GLM ANOVA	GLM ANOVA	GLM ANOVA/Contingency tables or Discriminant analysis	
Fixed	Both	ANCOVA Dummy regr	GLM ANCOVA	GLM ANCOVA	GLM ANCOVA or Discriminant analysis	
Random	Any kind	LMM	GLMM			
New sample	Any kind	Prediction			Classification	

Reflection Activity

Turn to your neighbour(s) and discuss:

- Do you know what kind of **data** you will be dealing with in your work?
- Do you have any idea what **association studies** or **methods** are relevant for your research?
- Are they mentioned in the table in the previous page?

Now, Let's Begin!

Lecture 1, Part B: Review of Basic Statistics

We'll start with basics of **probability theory** - the mathematical foundation for **uncertainty** and **statistical inference**.

Part 0: Foundations of Probability

Setting Up a Random Experiment

Before studying random variables, we need to define randomness formally.

A probability setup has three components:

1. **Sample space** Ω: all possible outcomes
   - *Discrete:* Coin flip: $\Omega = \{\text{H}, \text{T}\}$
   - *Continuous:* Adult height: $\Omega = [120, 250]$ cm
2. **Events** $\mathcal{F}$: set of subsets of outcomes we observe
   - *Discrete:* $\{\text{H}\}$ = "heads" or $\{\text{H}\}$ ="tail"
   - *Continuous:* $\{h: \text{ between 160 and 180}\}$ or $\{h: \text{ between 240 and 250}\}$
3. **Probabilities** P: assigns to events a number in $[0, 1]$
   - *Discrete:* $P(\text{H}) = 0.5$
   - *Continuous:* $P(160 \leq h \leq 180) = 0.68$ or $P(240 \leq h \leq 250) = 0.001$

Together: $(\Omega, \mathcal{F}, P)$ is a **probability space**.

Andrey Kolmogorov (1903–1987)

Figure 1: Soviet mathematician who revolutionized probability by providing rigorous mathematical foundations

- Established the **foundations of probability** (1933)
- Formalized the concept of probability space $(\Omega, \mathcal{F}, P)$.

Kolmogorov's Axioms:

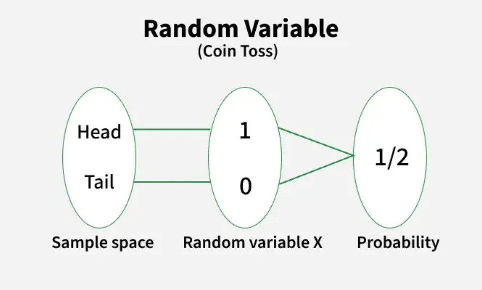
1. $P(\Omega) = 1$ (certainty has probability 1)
2. $P(A) \geq 0$ for all outcomes A (non-negativity)
3. Countable additivity: If $A_1, A_2, \ldots$ are disjoint events, then $P(\bigcup_i A_i) = \sum_i P(A_i)$

Part 1: Random Variables

What is a Random Variable?

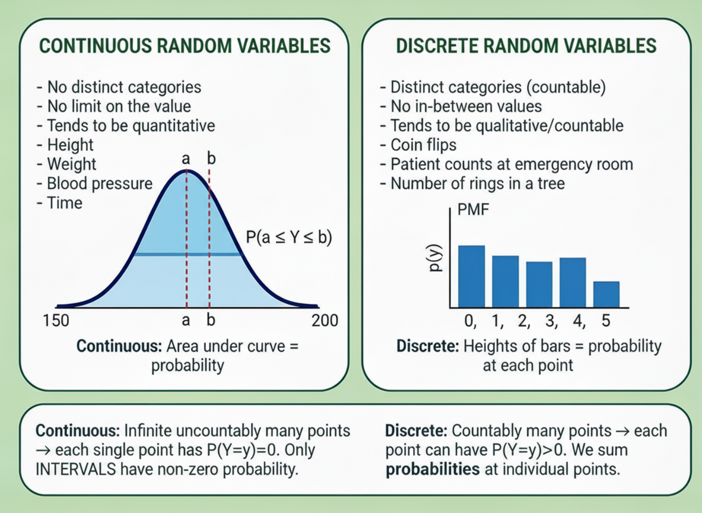
A **random variable** is a function that assigns a numeric value to each outcome in the sample space. Maps from experiment sample space to a real space.

Why? Numbers let us compute statistics: averages, spreads, and patterns.



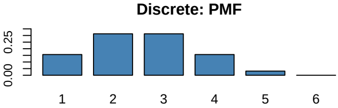
Example: Coin flip outcome $\rightarrow$ Number: - Heads $\rightarrow Y = 1$ - Tails $\rightarrow Y = 0$

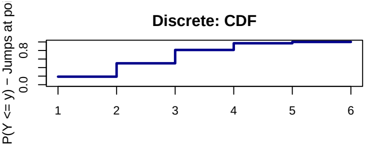
Discrete vs. Continuous Random Variables



Discrete Random Variables

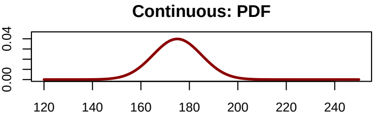
- **Countable** values (often integers)
- Examples: Coin flips, patient counts at emergency room, number of rings in a tree
- Use **probability mass function (PMF)** $p(y)$:
  $P(Y = y) = p(y)$ and $\sum_{\text{all } y} p(y) = 1$
- **Cumulative distribution function (CDF):**
  $F(y) = P(Y \leq y) = \sum_{x \leq y} p(x)$
- Also exist quantile functions, moment generating functions, characteristic functions (beyond this course).

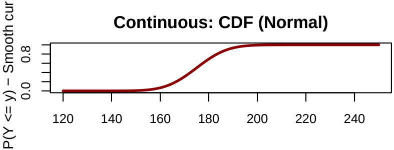




Continuous Random Variables

- Values in an **interval** (uncountably many)
- Examples: Height, weight, blood pressure, time
- Use **probability density function (PDF)** $f(y) \geq 0$
- Probabilities: $P(a \leq Y \leq b) = \int_a^b f(y)\,dy$ and $\int_{-\infty}^{\infty} f(y)\,dy = 1$
- **Cumulative distribution function (CDF):** $F(y) = P(Y \leq y) = \int_{-\infty}^{y} f(x)\,dx$





CRITICAL DIFFERENCE: Single Point Probability

CHECKPOINT: Why does P(Y = 172.5) = 0 for continuous height?

Property	Discrete	Continuous
$P(Y = y)$	**Can be > 0**	**Always $= 0$**
Example: Fair die	$P(Y = 3) = 1/6$	—
Example: Height	—	$P(\text{height} = 172.5) = 0$
How to find probability	Use $P(Y = y)$	Use $P(a \leq Y \leq b)$

Intuition: Continuous distributions have *uncountably many* points (infinity!), so each single point contributes zero probability. Only **intervals with area** have non-zero probability.

Expectation: Discrete Case

For a **discrete** Y with probabilities $p(y)$:

$$E(Y) = \mu = \sum_{\text{all } y} y \cdot p(y)$$

Fair die example ($\{1, 2, 3, 4, 5, 6\}$, each with $p(y) = 1/6$):

$$E(Y) = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + \cdots + 6 \cdot \frac{1}{6} = 3.5$$

Interpretation: Long-run average over many rolls.

CHECKPOINT: Understanding E(Y) for the Fair Die

Question: Why is $E(Y) = 3.5$, not 3 or 4? What does 3.5 mean?

- Is 3.5 a possible outcome when rolling a die? **NO**
- Does it mean we expect to see 3.5 on the next roll? **NO**
- Does it mean over 1,000 rolls, the average is *approximately* 3.5? **YES**

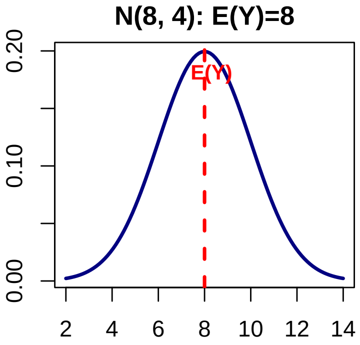
Key insight: $E(Y)$ is the **balance point** of the distribution. Imagine the die outcomes as weights on a seesaw—3.5 is where it balances!

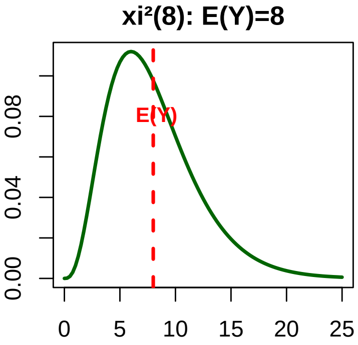
Expectation: Continuous Case

For a **continuous** Y with density $f(y)$:

$$E(Y) = \mu = \int_{-\infty}^{\infty} y\, f(y)\, dy$$

Interpretation: Center of mass (balance point) of the distribution.





Variance: Discrete Case

For a **discrete** Y:

$$Var(Y) = \sigma^2 = E((y-\mu)^2) = \sum_{\text{all } y} (y-\mu)^2 \cdot p(y)$$

Fair die with $\mu = 3.5$:

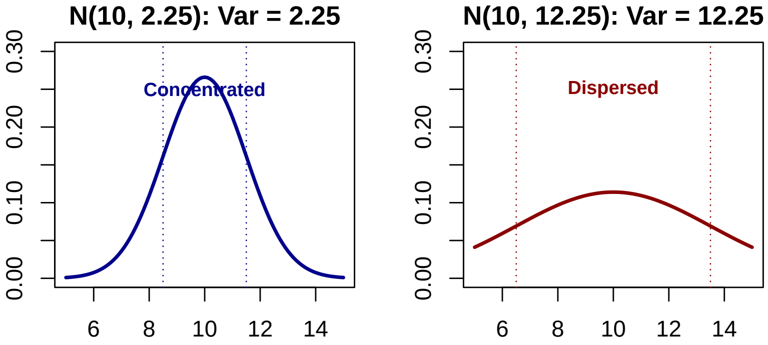
$$Var(Y) = (1-3.5)^2 \cdot \frac{1}{6} + \cdots + (6-3.5)^2 \cdot \frac{1}{6} \approx 2.92$$

Variance: Continuous Case

For a **continuous** Y:

$$Var(Y) = \sigma^2 = \int_{-\infty}^{\infty} (y - \mu)^2 f(y)\, dy$$

Standard deviation: $\sigma = \sqrt{Var(Y)}$ (same units as Y)



Larger variance = more spread!

Sample Mean and Standard Deviation

Given n observations $y_1, \ldots, y_n$:

Sample mean (estimates μ):

$$\bar{y} = \frac{1}{n} \sum_{t=1}^{n} y_t$$

Sample standard deviation (estimates σ):

$$s = \sqrt{\frac{1}{n-1} \sum_{t=1}^{n} (y_t - \bar{y})^2}$$

We divide by $(n-1)$ for unbiased variance estimation (**Bessel's correction**).

Part 5: Rules for Transformations

Linear Transformations: $E(cY)$ and $Var(cY)$

For constant c and random variable Y:

$$E(c) = c,$$
$$E(cY) = c \cdot E(Y)$$
$$Var(c) = 0,$$
$$Var(cY) = c^2 \cdot Var(Y)$$

Key insight: Variance scales with c^2. Doubling a variable multiplies variance by 4!

CHECKPOINT: can someone say how to prove $\mathrm{Var}(cY) = c^2\mathrm{Var}(Y)$, not $c\mathrm{Var}(Y)$?

Question: If we triple the values (multiply by 3), variance multiplies by 9. Why the square?

Sums of Independent Variables

For independent $Y_1, \ldots, Y_n$ with expectations μ_i and variances σ_i^2 and some constants c_i:

Expectation of sum:

$$E\left(\sum_{i=1}^{n} c_i Y_i\right) = \sum_{i=1}^{n} c_i \mu_i$$

Variance of sum (independent only):

$$Var\left(\sum_{i=1}^{n} c_i Y_i\right) = \sum_{i=1}^{n} c_i^2 \sigma_i^2$$

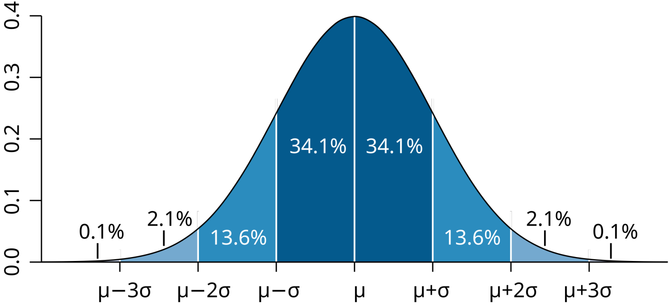
Part 6: Key Distributions

The Normal Distribution

Notation: $Y \sim N(\mu, \sigma^2)$

PDF: $f(y) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{-\frac{1}{2}\left(\frac{y-\mu}{\sigma}\right)^2\right\}$

Empirical Rule & Visual:



CHECKPOINT: SAT Scores and the 68-95 Rule

Application: SAT scores follow $N(\mu = 500, \sigma^2 = 100^2)$.

- What % score between **400-600**? (within 1 standard deviation)
- What % score between **600-700**? (1 to 2 standard deviations away from mean)

CHECKPOINT: SAT Scores and the 68-95 Rule

Application: SAT scores follow $N(\mu = 500, \sigma = 100)$.

- What % score between **400-600**? (within 1 standard deviation)
  - **Answer: 68.2%**
- What % score between **600-700**? (2 standard deviations away from mean)
  - **Answer: 13.6%**

Why this matters: Standardized tests are designed using normal distributions. Knowing the 68-95 rule helps interpret scores!

Carl Friedrich Gauss (1777–1855)

The "Prince of Mathematicians" and namesake of the normal distribution.

Figure 2: German mathematician with groundbreaking contributions to statistics (**error theory** back then), calculus, algebra and number theory, and astronomy

Linear Combinations of Normal Variables

If $Y_1 \sim N(\mu_1, \sigma_1^2)$ and $Y_2 \sim N(\mu_2, \sigma_2^2)$ are *independent*:

$$aY_1 + bY_2 \sim N(a\mu_1 + b\mu_2,\ a^2\sigma_1^2 + b^2\sigma_2^2)$$

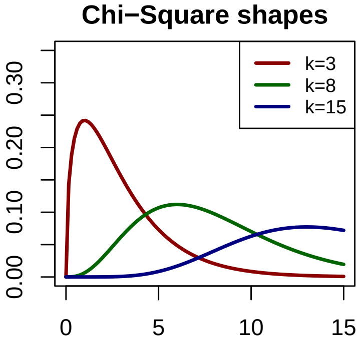
This is **fundamental** for inference about sample means and regression.

The Chi-Square Distribution

Notation: $Y \sim \chi^2_k$ where k = degrees of freedom

Key derivation: $Z^2 \sim \chi^2_1$ if $Z \sim N(0,1)$; additive property: $Y = \sum_{i=1}^{k} Z_i^2 \sim \chi^2_k$

Properties: $E(Y) = k$, $Var(Y) = 2k$

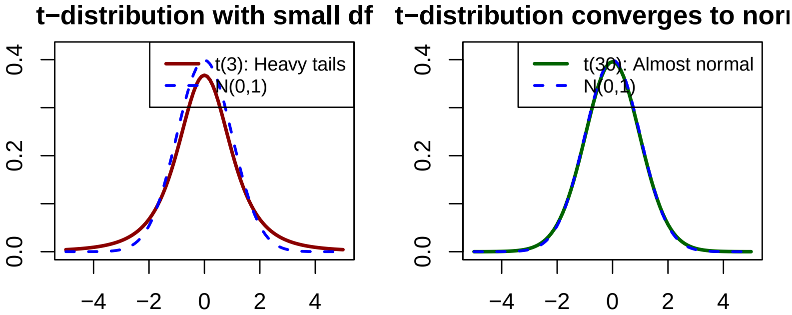


The Student's t Distribution

Definition: For independent $Z \sim N(0,1)$ and $Y \sim \chi^2_k$:

$$T = \frac{Z}{\sqrt{Y/k}} \sim t_k$$

$$E(T) = 0 \quad \text{and} \quad Var(T) = \frac{k}{k-2}$$



The t Distribution: Practical Use

Test statistic for population mean:

If $y_1, \ldots, y_n$ i.i.d. $N(\mu, \sigma^2)$ and $S^2 = \frac{1}{n-1} \sum_{t=1}^{n} (y_t - \bar{y})^2$:

$$T = \frac{\bar{y} - \mu}{\sqrt{S^2/n}} \sim t_{n-1}$$

This is the **one-sample** t test statistic.

William Sealy Gosset (1876–1937): "Student"

Figure 3: English scientist working as a chemist for Guinness Brewery. Developed the t distribution to handle small samples when variance is unknown (1908)

- Faced a practical problem: small sample inference for quality control
- Published under the pseudonym **"Student"** because Guinness forbade employee publications

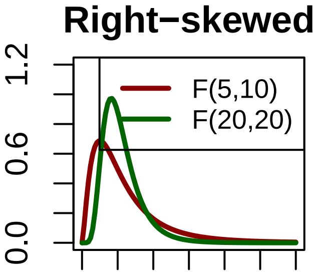
The F Distribution

Notation: $Y \sim F_{u,v}$ where u, v = degrees of freedom

Definition: For independent χ^2_u and χ^2_v:

$$F = \frac{\chi^2_u/u}{\chi^2_v/v} \sim F_{u,v}$$

$$E(Y) = \frac{v}{v-2}, v > 2 \quad Var(Y) = \frac{2v^2(u+v-2)}{u(v-2)^2(v-4)} v > 4$$



Ronald Aylmer Fisher (1890–1962)

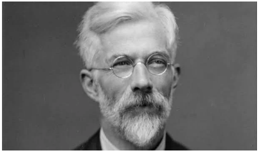


Figure 4: British statistician and geneticist with revolutionary impact on statistics and evolutionary biology. The founder of modern statistics and namesake of the F-distribution.

- Invented the **F distribution**, **ANOVA** (analysis of variance), and **maximum likelihood estimation**
- Pioneered **experimental design** principles (randomization, blocking, replication)
- Won the prestigious **Weldon Medal** for his contributions to statistical science

Legacy: Fisher's work transformed statistics from a descriptive tool to a framework for scientific inference.

Many more distributions of course